23.10.2026 items review

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1 Status of the 08.10 items

1. **Tune the weights of quality, continuity and leakage. Done.**
2. **Capacity in the QUBO. Done.** Kept outside.
3. **Build and evaluate the QUBO. Built, evaluation in progress.** One QUBO per antenna and timeslot (262 in the reference run), at most 20 variables each, all off-diagonal terms ≥ 0 as the neutral atom embedding requires. An exhaustive check over every state confirms on all 262 that the QUBO energy equals minus the objective and that no state with a hard pair is optimal. The comparison against the interaction matrix continues next week with the checks suggested in the meeting (Section 2).
4. **Try the QUBO solver on simpler, smaller problems. Open.** `qubo-solver` installed and running.
5. **Weight precision study ($\pm 5\%$). Open.**
6. **Effect of the prune and the repair on solution quality. Partly done.** The prune is measured (item 2). The effect of the coverage repair is still open.
7. ****EXTRA** Report. Done** First draft of the introduction and the state of the art sections.

2 Context

1. **How to check the QUBO against the interaction matrix.**
   - Global comparison of the QUBO matrix Q against the interaction matrix U of the register.
   - Compare the order of the coefficients: the strongest couplings of Q should remain the strongest in U.
   - Solve the QUBO built from U and apply its solution to the original QUBO, to check whether it is still good.

2. **Preprocessing and fixed variables.** When the preprocessing fixes a variable to 1, its atom is removed, and its neighbours lose the coupling that kept them out: the neighbourhood is unlocked. Classically the reduction is exact, since `qubo-solver` adds each removed coupling $2Q_{uv}$ to the neighbour's diagonal term. On the hardware, that shift only exists if it reaches the neighbouring atoms. Options:

(a) Keep variables fixed to 1 in the register (do not remove the atom), so the coupling is physically present.

(a) Pass the information on to the pulse design, so that the neighbouring atoms receive the shift through local detuning.

This affects the quantum hardware path only, not annealing or the other classical approaches, which work directly on the reduced matrix.

3 Items

1. Check the QUBOs against the interaction matrix with the three checks above.

2. Check the preprocessing: how often variables are fixed to 1 in our QUBOs and how many neighbours are affected, consider options (a) and (a).

3. Get used to the solver on example problems.

4 Benchmarking plan

Same scenario and random seed used on every solver; every schedule is checked with the same validity rules.

- Greedy oracle: the baseline.
- QUBO/MIS, solved classically (checks the QUBO formulation agrees with Greedy).
- Pulser based quantum sampling on the emulator/QPU.

Metrics: gap to the exact optimum per QUBO; total spectral efficiency, coverage (must be 100%), worst, 10th percentile and median user rate, Jain index, handovers and new users served, runtime against QUBO size.